



2.2 Quadratic models

Functions · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A quadratic model $y=ax^2+bx+c$ produces a parabola; its vertex, at $x=-b/(2a)$, is a maximum ($a<0$) or minimum ($a>0$) point. Quadratics model contexts with one turning point, such as projectile height, revenue/profit against price, or area optimisation. Roots (where $y=0$) can be found by factorisation, the quadratic formula, or technology, and give break-even points or landing times.

Worked example: follow the reasoning

A ball's height above ground, $h(t)$ metres, t seconds after being thrown is $h(t)=-5t^2+20t+1.5$. Find the maximum height and the time (to 2 dp) at which the ball lands.

Solution: Vertex $t=-20/(2 \times -5)=2$ s; $h(2)=-20+40+1.5=21.5$ m (maximum height). Landing: $-5t^2+20t+1.5=0 \rightarrow 5t^2-20t-1.5=0 \rightarrow t=(20+\sqrt{(400+30)})/10=(20+\sqrt{430})/10 \approx 4.07$ s.

Quick check · multiple choice

What is the vertex of $y=2(x-3)^2+5$?

- A (3,5)
- B (-3,5)
- C (3,-5)
- D (5,3)

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A company's weekly profit (\$) from selling x units of a product is modelled by [6]
 $P(x)=-2x^2+180x-1500$. (a) Find the number of units that maximizes profit, and the maximum profit. (b) Find the values of x for which the company breaks even ($P(x)=0$), to the nearest whole unit.

Final answer: _____



Answers and explanation

Question 1 [6]

(a) Vertex $x = -180/(2 \times -2) = 45$. $P(45) = -2(2025) + 180(45) - 1500 = -4050 + 8100 - 1500 = \2550 maximum profit at 45 units. (b) $-2x^2 + 180x - 1500 = 0 \rightarrow x^2 - 90x + 750 = 0 \rightarrow x = (90 \pm \sqrt{5100})/2 \approx (90 \pm 71.41)/2$, giving $x \approx 9$ or $x \approx 81$.

Quick check answer

A. In vertex form $y = a(x-h)^2 + k$, the vertex is $(h, k) = (3, 5)$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.